0780 PHYSICS 1

GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2024 **ADVANCED LEVEL**

Centre Number	
Centre Name	
Candidate Number	
Candidate Name	

Mobile phones are NOT allowed in the examination room.

MULTIPLE CHOICE QUESTION PAPER

Duration: One and Half Hours

INSTRUCTIONS TO CANDIDATES

Read the following instructions carefully before you start answering the questions in this paper. Make sure you have a soft HB pencil and an eraser for this examination.

1. USE A SOFT HB PENCIL THROUGHOUT THE EXAMINATION.
2. DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Before the examination begins:

3. Check that this question booklet is headed "**Advanced Level – 0780 PHYSICS 1**"
4. Fill in the information required in the spaces above.
5. Fill in the information required in the spaces provided on the answer sheet using your HB pencil: **Candidate Name, Exam Session, Subject Code and Candidate Identification Number.** Take care that you do not crease or fold the answer sheet or make any marks on it other than those asked for in these instructions.

How to answer the questions in this examination

6. Answer **ALL** the **50** questions in this Examination. All questions carry equal marks.
7. Calculators are allowed.
8. Each question has FOUR suggested answers: **A, B, C** and **D**. Decide which answer is correct. Find the number of the question on the Answer Sheet and draw a horizontal line across the letter to join the square brackets for the answer you have chosen.

For example, if **C** is your correct answer, mark **C** as shown below:

[A] [B] [~~C~~] [D]

9. Mark only one answer for each question. If you mark more than one answer, you will score a zero for that question. If you change your mind about an answer, erase the first mark carefully, then mark your new answer.
10. Avoid spending too much time on any one question. If you find a question difficult, move on to the next question. You can come back to this question later.
11. Do all rough work in this booklet, using, where necessary, the blank spaces in the question booklet.
12. **At the end of the examination, the invigilator shall collect the answer sheet first and then the question booklet. DO NOT ATTEMPT TO LEAVE THE EXAMINATION HALL WITH IT.**

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SECTION I (Thirty-five questions)

Question: 1- 35

Directions: Each of the questions or incomplete statements in this section is followed by four suggested answers. Select the best answer in each case

1. The base unit of magnetic field intensity is
   A $kg\ m^2\ s^{-2}\ A^{-1}$
   B $kg\ s^{-2}\ A$
   C $kg\ s^{-2}\ A^{-1}$
   D $kg\ s^{-1}\ A^{-1}$

2. The back of the hand feels cold when some alcohol is dropped on it because
   A alcohol is at a lower temperature than the atmosphere
   B alcohol has a larger heat capacity than your body
   C alcohol has a lower heat capacity than your body
   D alcohol extracts latent heat of vaporization from the hand.

3. Which of the following pairs both make use of critical damping?
   A Simple pendulum and galvanometer
   B Galvanometer and shock absorber in cars
   C House foundations and shock absorbers in cars
   D Simple pendulum and bridge construction

4. The area under a force (F) against time (t) graph for a moving object is the;
   A change in linear momentum of the object
   B total distance covered by the object
   C acceleration of the object
   D work done by the object

5. The resultant force acting on object O in figure I is;

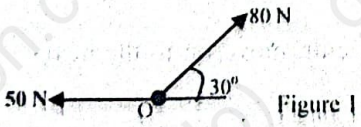


Figure I

   A 44 N North-East of O
   B 30 N North-East of O
   C 44 N West of O
   D 40 N North of O

6. The strings of a nylon guitar of cross-sectional area $0.5 \times 10^{-6}\ m^2$ store elastic potential energy of 0.8 J when stretched by 2.0 *cm*. If Young's modulus for nylon is $2 \times 10^9\ N\ m^{-2}$, then the length of the string is:
   A 25.0 m
   B 50.0 cm
   C 25.0 cm
   D 80.0 cm

7. Platinum resistance thermometers are quite sensitive because;
   A platinum has a high temperature coefficient of resistance
   B the thermometers have relatively low heat capacities
   C platinum has a high melting point
   D the thermometers have relatively high heat capacities

8. A sample of gas has a root mean square (rms) speed of $2.2 \times 10^3\ m\ s^{-1}$ at STP. What is the value of its rms speed at 80 ^{0}C and pressure of one atmosphere?
   A $2.8 \times 10^3\ m\ s^{-1}$
   B $2.5 \times 10^3\ m\ s^{-1}$
   C $2.0 \times 10^4\ m\ s^{-1}$
   D $1.3 \times 10^3\ m\ s^{-1}$

9. Figure 2 shows a graph of stopping potential V_s plotted against frequency, f, for nichrome

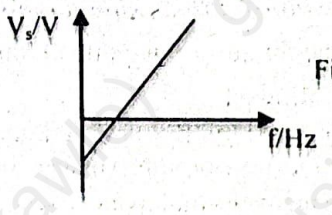


Figure 2

Which of the following is **CORRECT**?
   A The product of gradient of the graph, the value of the intercept on the horizontal axis and the electronic charge, is the work function for nichrome.
   B The value of the intercept on the vertical axis is the work function for nichrome.
   C The product of gradient of the graph and the value of the intercept on the horizontal axis, is the work function for nichrome.
   D The gradient of the graph is the Planck's constant

10. Figure 3a shows a spring of spring constant K extended by a mass m. In figure 3b two identical springs connected as shown are extended by the same mass m attached to the middle of a short connecting bar

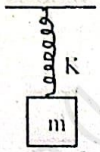


Figure 3a

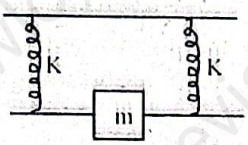


Figure 3b

If both systems are set into oscillation, which of the following is true?

A Mass, m, oscillates faster in figure 3b than in figure 3a
B Mass, m, has a longer period of oscillation in figure 3b than in 3a
C The combined spring constant for the system in 3b is half that in 3a
D Mass, m, extends the combined springs in 3b twice as much as the single spring in 3a

11. Given that the decay constant for Co-60 is 0.132. Its half-life is:

A 15.15 years
B 26.40 years
C 5.25 years
D 9.15 years

12. Two parallel plates each of area 0.04 m^2 are placed 5 mm apart in a vacuum as shown in figure 4,

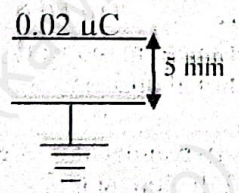


Figure 4

The potential of the upper plate is;

A 225 V
B 240 V
C 120 V
D 282 V

13. A force of 1.6×10^{-6} N acts between two parallel wires in vacuum as shown in figure 5. Given that the wires each carry a current I.

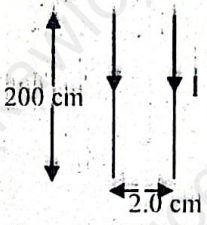


Figure 5

The value of I is;

A 0.08 A
B 0.3 A
C 0.2 A
D 4.0 A

14. Figure 6 represents a cyclic thermodynamic process. If the heat supplied is dQ and the change in internal energy is dU, the quantity of heat supplied to the system is given by:

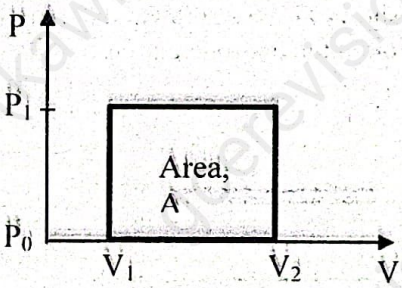


Figure 6

A dQ=dU+A
B dQ=dU+P_1A
C dQ=dU-PA
D dQ=du-A(V_2-V_1)

15. Figures 7 a and 7 b show the arrangement of bulbs each with resistance as shown.

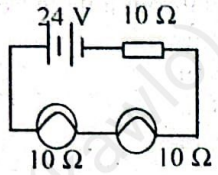


Figure 7 a

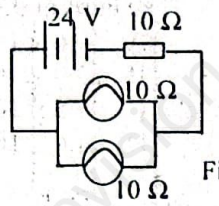


Figure 7 b

Which of the following is true?

A The brightness of the bulbs in figure 7b is the same but more than those in 7a
B The brightness of all four bulbs is the same but battery in figure 7b will run down faster
C The brightness of the bulbs in figure 7a is same but more than that in 7b
D The brightness of all four bulbs is the same and both batteries will last for the same length of time.

16. Figure 8 shows the energy-level diagram for the electron in the hydrogen atom.

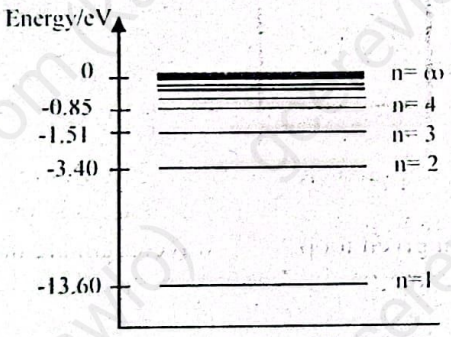


Figure 8

When the electron in the hydrogen atom makes a transition from energy level n=4 to the ground state n = 1, it emits light of frequency

A 2.1×10^{14} Hz
B 1.6×10^{15} Hz
C 3.1×10^{15} Hz
D 3.3×10^{15} Hz

17. N-type semiconductors are more widely applicable in electronic devices than P-type because

A the net charge in the N-type is electrons
B N-type are easier to manufacture than P-type
C the N-type are more compact than the P-type.
D electrons move faster than holes

18. An object A, of mass 3.5 kg is accelerated from rest by a net force of 5 N in 2.0 s. Another object, B,of mass 2.8 kg is accelerated from rest by the same force in 1.6 s. The ratio of the gain in kinetic energy of A to that of B is

A 1:1
B 1.25:1
C 2:1.6
D 2:1

19. The time taken for an object in figure 9 to move from point O to P is;

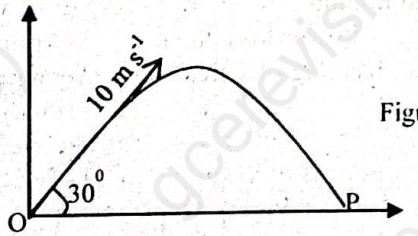


Figure 9

A 0.5 s
B 2.0 s
C 1.0 s
D 1.7 s

20. The molar heat capacity of a gas at constant pressure is always greater than that at constant volume because;

A at constant pressure, work is done against external pressure in expanding the gas
B the molar gas constant makes up the difference between the two heat capacities
C at constant volume, the gas does not require much energy to raise the temperature
D at constant pressure, no work is done against external pressure

21. At a given instant, five molecules of gas have speeds 100, 200, 250, 300, 600 m s^{-1} respectively. The root mean square speed of the gas molecules is:

A 426 m s^{-1}
B 335 m s^{-1}
C 247 m s^{-1}
D 329 m s^{-1}

22. A sonometer wire of material of density 5×10^3 kg m^{-3} is stretched such that transverse waves are transmitted along it at a speed of 160 m s^{-1}. The tension per unit area of cross section of the stretched wire is;

A 8.0×10^5 N m^{-2}
B 9.8×10^8 N m^{-2}
C 1.3×10^8 N m^{-2}

23. Sodium light of wavelength 600 nm falls normally on a diffraction grating. If the angle between the first order image and the normal is 17.5^0. The number of lines per millimetre ruled on the grating is;
A 200
B 501
C 5.01×10^5
D 2.0×10^3

24. Which of the following methods of power losses in a transformer corresponds to the method of minimizing it?

	Power loss due to	Minimized by
A	Heating of wires	Wrapping the primary coil round the secondary
B	Poor flux linkage	Using thick copper wires
C	Eddy currents	Laminating the transformer core
D	Continuous magnetization and demagnetization of core	Using thick wires

25. An object is placed 15.0 cm from a converging lens of focal length 10.0 cm. The image formed is;
A on the same side of lens as the object, magnified, virtual
B half the height of the object, inverted, erect
C on the same side of lens as the object, inverted, real
D twice the height of object, inverted, real

26. Figure 10 is a graph of the variation of the current through a 0.04 kΩ resistor in circuit with time.

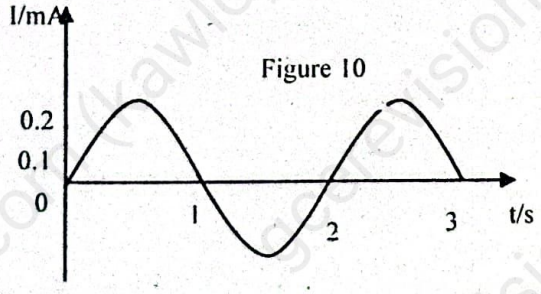


Figure 10

The root mean square voltage across the resistor in millivolts is;
A 8.0
B 5.7
C 0.1
D 2.0

27. Electrons drift through a section of a piece of copper wire containing 10^{29} electrons per unit volume with a velocity of $2.9 \times 10^{-4}\ m\,s^{-1}$. The current density through the wire is;
A $4.6 \times 10^6\ A\ m^{-2}$
B $2.9 \times 10^{25}\ A\ m^{-2}$
C $5.5 \times 10^{13}\ A\ m^{-2}$
D $4.6 \times 10^{13}\ A\ m^{-2}$

28. The sun emits electromagnetic radiation at the rate of $2.5 \times 10^{26}\ W$. The rate at which the sun is losing mass every second is;
A 8.3×10^{17} kg
B 5.6×10^9 kg
C 2.8×10^9 kg
D 1.4×10^8 kg

29. Which of the following is **NOT** exhibited by longitudinal waves?
A Diffraction
B Refraction
C Interference
D Polarization

30. Which of the following experimental results of the photoelectric effect can be explained by the classical or wave theory?
A The kinetic energy of emitted electrons depending on frequency of incident radiation
B Number of photoelectrons emitted per unit time increases with light intensity
C Maximum speed of emitted electrons is independent of the intensity of the incident radiation?
D The instantaneous emission of electrons from a metal surface even with faint incident radiation

31. The impedance of the circuit in figure 11 is;

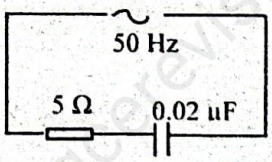


Figure 11

A 64.5 Ω
B 1.59×10^5 Ω
C 8.03 Ω
D 2.5×10^5Ω

32. For a fixed mass of gas undergoing an adiabatic change, which of the following is correct?
A The process should be carried out very fast.
B There is no change in internal energy of the system.
C The walls of the containing vessel should be thin.
D The initial temperature of the system must necessarily be equal to its final temperature.

33.

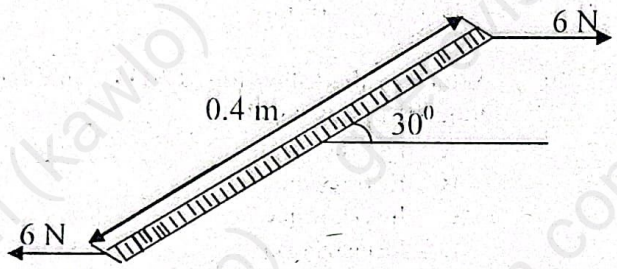


Figure12

Figure 12 shows a couple.
The magnitude of the moment of the couple is
A 2.4 N m
B 4.8 N m
C 2.1 N m
D 1.2 N m

34. Which of the following about capacitors is **NOT** correct?
A They allow alternating current to pass through them
B They store charge
C They store energy
D Direct current flows through them only in one direction.

35. Which of the following explains why the electrical resistance of metals increases with increase in temperature?
A More positive ions are produced at a higher temperature than conduction electrons.
B Electrons do not move fast at higher temperature.
C Amplitudes of vibration of positive ions increase with temperature, impeding movement of conduction electrons.
D Conduction electrons are captured by positive ions at higher temperature.

SECTION II (Ten questions)
Multiple Selection

Questions 36- 45 Each question or incomplete statement in this section is followed by three numbered responses. One or two of the responses may be correct. CHOOSE:
A. If 1 and 2 are correct
B. If 2 and 3 are correct
C. If 1 only is correct
D. If 3 only is correct

Directions Summarised			
A	B	C	D
1,2 only	**2, 3 only**	**1 only**	**3 only**

36. The force between two small conducting spheres carrying identical charges and placed 4 mm apart in acuum is 2.0×10^{-3} N. Which of the following is/are correct?
1 The force is attractive
2 The charge on one of the spheres is 1.9 ηC
3 The force is repulsive

37. Which of the following is/are a difference(s) between stationary and progressive waves?

	Stationary waves	Progressive waves
1	Store energy	Transmit energy
2	Particles within a given loop vibrate in phase	Particles within a wavelength are not in phase
3	The wave profile advances with the wave speed	The wave profile does not advance with the wave speed

38. Figure 13 shows the mean distribution of kinetic energy, ke, among a number of molecules N, of a gas at constant temperature T_1.

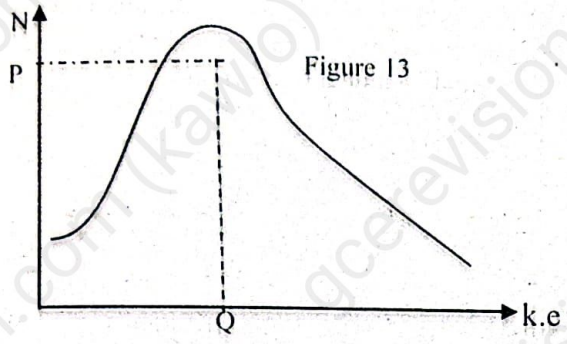


Figure 13

For temperature $T_2 < T_1$, the value of
1 P increases
2 Q decreases
3 P stays constant

39. Passengers in a moving car jerk-forward when the driver suddenly apply brakes. This is a consequence of ;
1 Newton's third law of motion
2 law of conservation of energy
3 Newton's first law of motion

40. Which of the following statements is/are true about fields?
1 Magnetic fields are produced by moving charges
2 A gravitational field is due only to the presence of a charge
3 An electric field is produced only by a fast moving charge

41. A mole of an ideal gas contained in a 0.025 cubic metre vessel is subjected to a pressure of 10^5 Pa. Which of the following is/are correct?
1 If the pressure of the gas in the same containing vessel is doubled, the temperature of the gas will decrease by half
2 The temperature of the gas is 25 ^{0}C
3 The mean kinetic energy of the gas molecule is 6.2×10^{-21} J

42. The second law of thermodynamics helps one to understand that;
1 net heat flows only in a particular direction in the universe.
2 the entropy of the universe changes with time.
3 heat does not flow from a cold to hot object naturally.

43. X-rays are produced when:
1 fast moving electrons strike a metal target and are brought to rest suddenly
2 electrons in discrete higher energy levels in an atom make a transition to the lower energy levels.
3 a metal surface is illuminated with light of appropriate frequency.

44. Which of the following deduction(s) from the alpha-particle scattering experiment ties with the conclusion?

	Deduction	Conclusion
1	A few α-particles were deflected through large angles	The positively charged core occupies a very tiny part of the atom
2	A very tiny proportion of the α-particles bounce back	The atom consists of a positively charged core
3	Most of the α-particles passed through the foil undeflected	The atom consists of mostly empty space

45. Figure 14 shows the magnetic field that results when a piece of iron is placed between unlike magnetic poles.

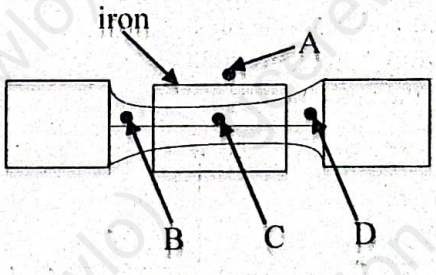


figure 14

At which point or points is the magnetic field strength strongest?

1 Points B and D
2 Point A
3 Point C

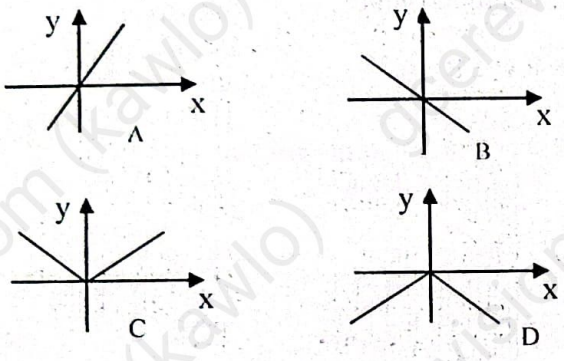
SECTION III (five questions)

Questions 46 - 50

Direction: Each of the questions (46-50) has four sets of graphs A - D. Which of the graphs in each question best fits the relationship between x and y?

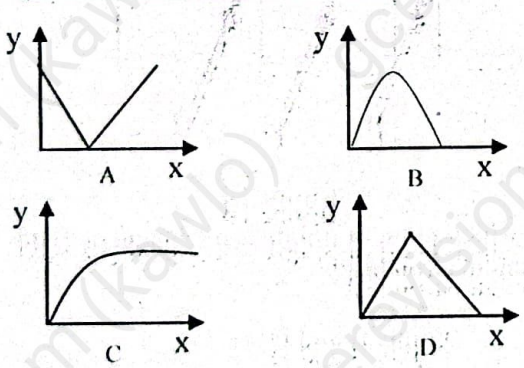
46.

y	x
Displacement of a body undergoing SHM	Acceleration



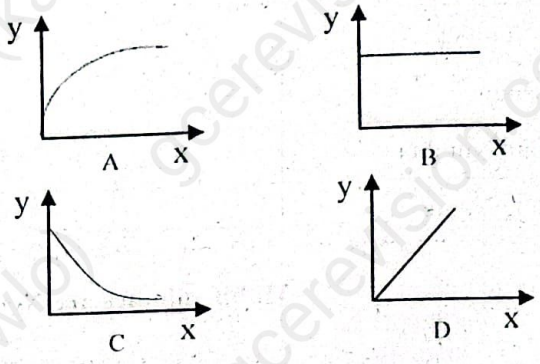
47.

y	x
The velocity of a train that takes off from rest, at one station and accelerates uniformly before stopping at the next station.	Time



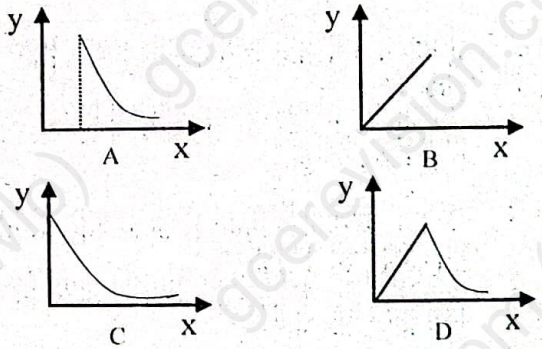
48.

y	x
Voltage across a diode	Current through the diode



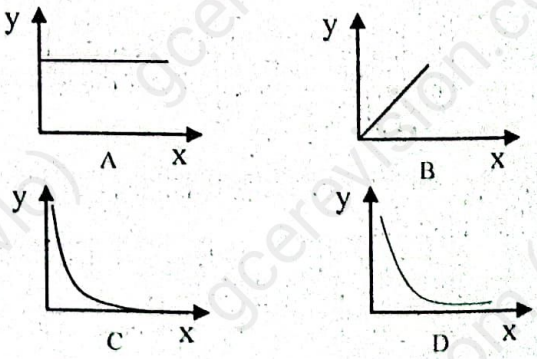
49.

y	x
The square of the velocity of a satellite orbiting the earth	Radius of orbit of the satellite



50.

y	x
Mass of a rocket moving with constant linear momentum	Velocity of the rocket



STOP

GO BACK AND CHECK YOUR WORK